

Identify common factors

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find the highest common factor of 84 and 126. Copy or complete the first step.

2. List the common factors of 36 and 60.

3. Find the HCF of 96 and 144.

4. Miro says: Miro chooses the largest factor from either list rather than the largest shared factor. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the highest common factor of 84 and 126. Copy or complete the first step.
Answer 42.
2. List the common factors of 36 and 60.
Answer 1, 2, 3, 4, 6 and 12.
3. Find the HCF of 96 and 144.
Answer 48.
4. Miro says: Miro chooses the largest factor from either list rather than the largest shared factor. Is this correct? Explain.
Answer A common factor must divide both numbers exactly.

Identify common factors

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find the highest common factor of 84 and 126.

6. Check this result using an inverse, estimate or known fact: Find the HCF of 96 and 144.

2. List the common factors of 36 and 60.

7. Two numbers have HCF 18. Give a possible pair greater than 50.

3. Find the HCF of 96 and 144.

8. Identify and correct this misconception: Miro chooses the largest factor from either list rather than the largest shared factor.

4. Solve this independently and show every stage: Find the highest common factor of 84 and 126.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: List the common factors of 36 and 60.

10. Write and solve a similar question to: Find the HCF of 96 and 144.

Teacher answers and checking prompts

1. Find the highest common factor of 84 and 126.
Answer 42.
2. List the common factors of 36 and 60.
Answer 1, 2, 3, 4, 6 and 12.
3. Find the HCF of 96 and 144.
Answer 48.
4. Solve this independently and show every stage: Find the highest common factor of 84 and 126.
Answer 42.
5. Use a second method or representation for: List the common factors of 36 and 60.
Answer Expected result: 1, 2, 3, 4, 6 and 12. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Find the HCF of 96 and 144.
Answer Expected result: 48. A valid check must be shown.
7. Two numbers have HCF 18. Give a possible pair greater than 50.
Answer 54 and 72 is one example.
8. Identify and correct this misconception: Miro chooses the largest factor from either list rather than the largest shared factor.
Answer A common factor must divide both numbers exactly.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Find the HCF of 96 and 144.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Identify common factors

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Two numbers have HCF 18. Give a possible pair greater than 50.

6. Create a new example that tests the same idea as: Find the HCF of 96 and 144.

2. Prove the answer to this question, rather than only calculating it: Find the highest common factor of 84 and 126.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: List the common factors of 36 and 60.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 48.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro chooses the largest factor from either list rather than the largest shared factor.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Two numbers have HCF 18. Give a possible pair greater than 50.
Answer 54 and 72 is one example.
2. Prove the answer to this question, rather than only calculating it: Find the highest common factor of 84 and 126.
Answer Expected result: 42. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: List the common factors of 36 and 60.
Answer Expected result: 1, 2, 3, 4, 6 and 12. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 48.
Answer One valid reconstruction is: Find the HCF of 96 and 144. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro chooses the largest factor from either list rather than the largest shared factor.
Answer A common factor must divide both numbers exactly.
6. Create a new example that tests the same idea as: Find the HCF of 96 and 144.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.